Retail Sentiment Intelligence

Real-Time Brand Health Monitoring via Reddit NLP Pipeline

BITS ZG628T · Dissertation — Post Mid-Semester Progress Presentation

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Dissertation work carried out at **Walmart Global Tech, Bengaluru***Supervisor:* **Mr. Varunendra Pratap Singh** — Principal Software Engineer, Walmart Global Tech

What We're Building & Where We Are

Project objective + progress checkpoint · **Part A delivered at mid-sem** · **Part B is today**

An **automated, trust-aware pipeline** that tracks Walmart-related conversations on Reddit, uses **LLMs** to extract sentiment, aspects & emerging themes, and surfaces them in a **human-reviewed dashboard** so Ops · PR · Product teams can respond at social speed.

👂 01 · Listen
25 Walmart / retail subreddits · hourly · Arctic Shift API.

🧠 02 · Analyze
ModernBERT · DeBERTa-v3 · Gemma 3 vision · Trust score.

👁️ 03 · Review
Analysts validate, correct & prioritize in a Kanban board.

🚀 04 · Act
Route to Ops · PR · Product via P1 / P2 / P3 with SLAs.

✅ Part A · Mid-Semester
DELIVERED · DEMOED TO COMMITTEE

- ✓ End-to-end **6-stage pipeline** (Ingest → Alert)
- ✓ **Vision pipeline** — Gemma 3 4B on image-only posts
- ✓ **Trust score** — Metadata · Dedup · LLM credibility
- ✓ **ModernBERT** fine-tune — Macro F1 **0.76** (vs RoBERTa 0.63)
- ✓ **Real-time dashboard** — 22 API endpoints + WebSocket
- ✓ **Notification system** — P1/P2/P3 (Email · Slack · Teams)

🟡 Part B · Post Mid-Semester
IN PROGRESS · PRESENTING TODAY

- ☐ **Review & Validate** — Human-in-the-Loop workflow
- ☐ **Post Explorer** — Multi-facet filtering & search
- ☐ **Kanban Lifecycle** — Triaged → In-Review → Resolved
- ☐ **Insights & Competitor** — Cross-brand comparison
- ☐ **Smart Reply Composer** — Dual-draft (Rule + LLM)
- ☐ **Learning loop** — Feedback → future retraining

Overall Dissertation Progress

Phase 1 of 2 · **~55% complete**

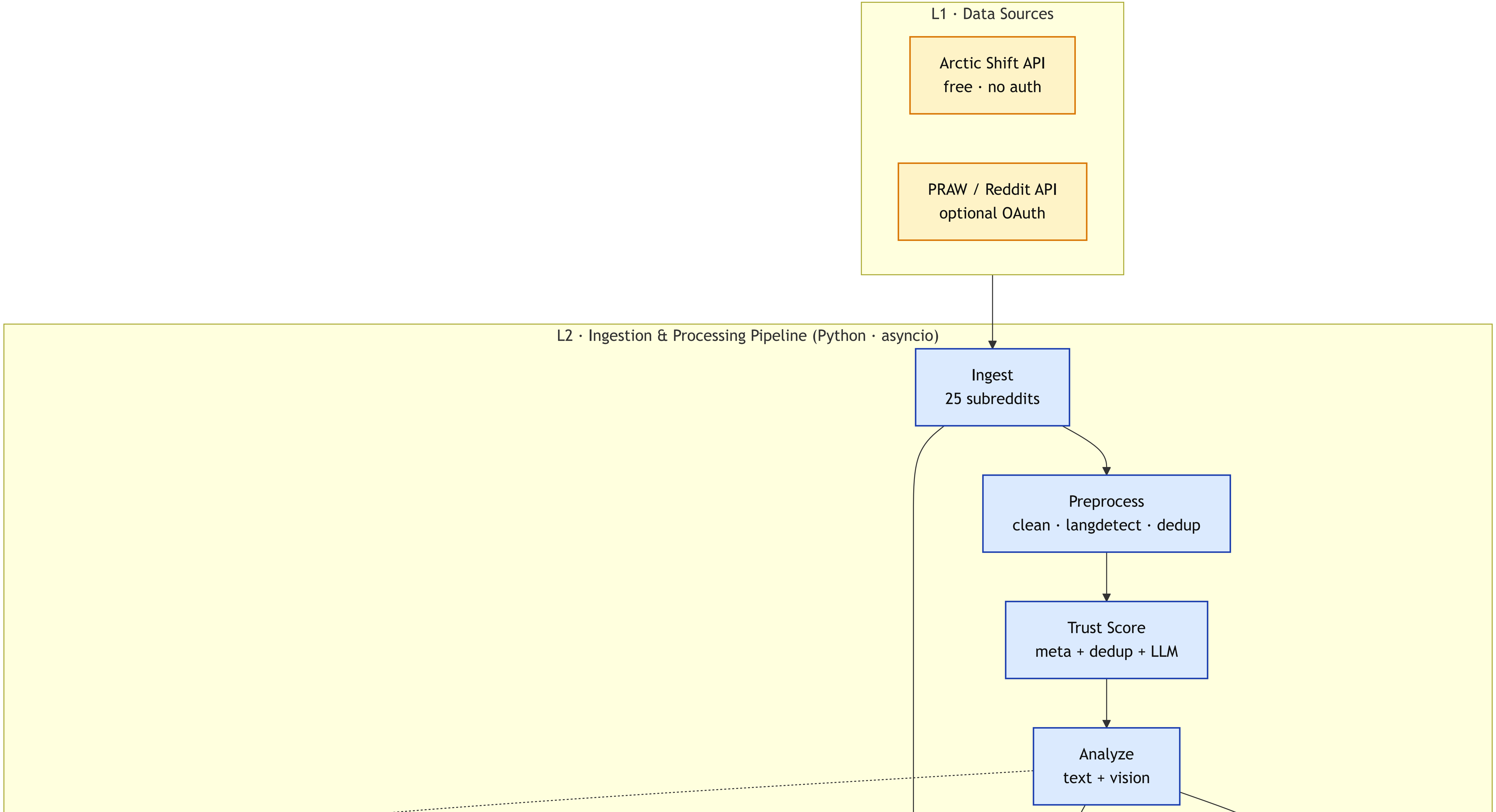
Start

↑ Mid-sem checkpoint (55%)

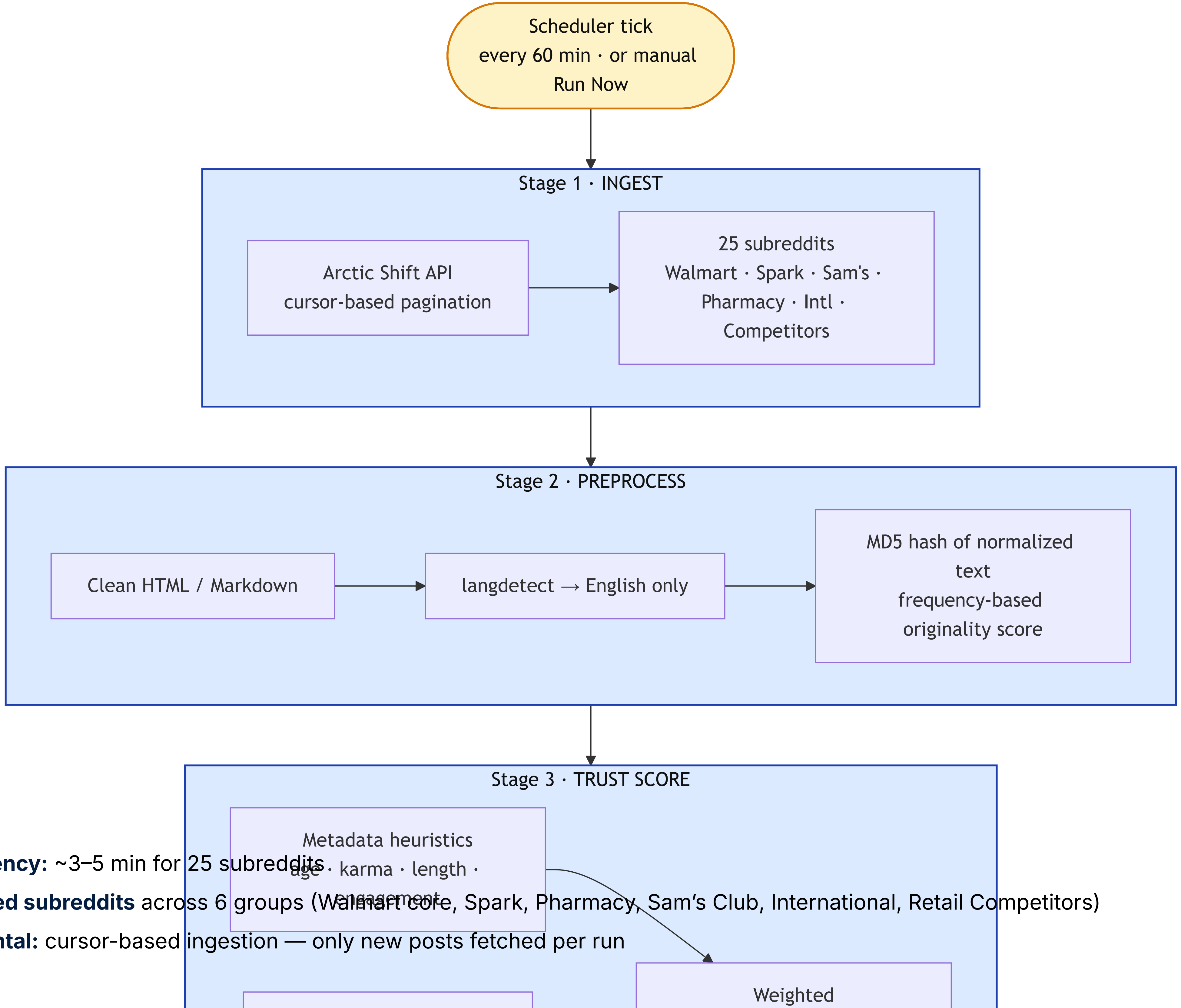
Final defence

High-Level System Architecture

Layered view — external sources → pipeline core → AI runtime → storage → serving → clients

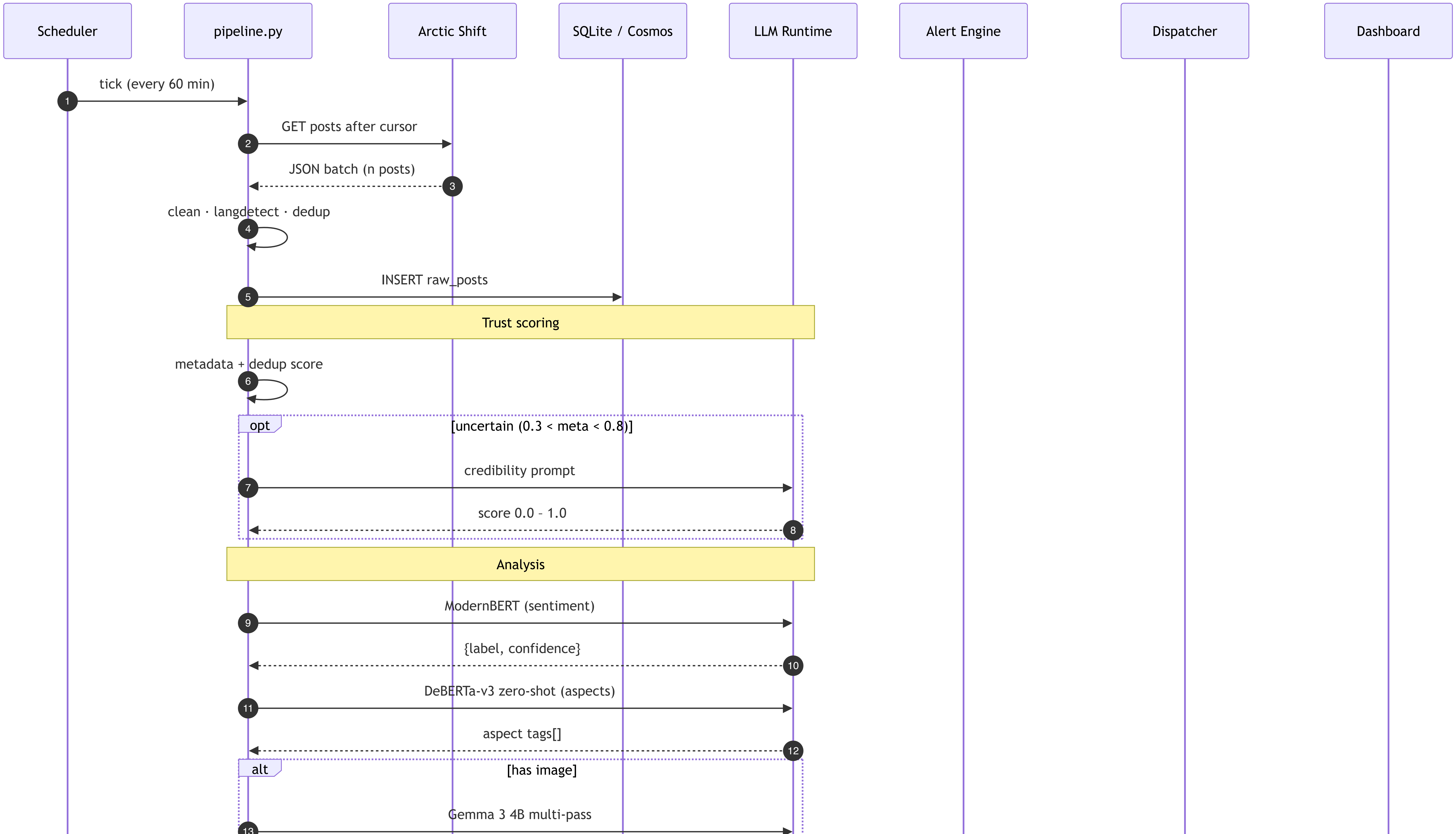


6-Stage Pipeline — Detailed Flow



- **Total latency:** ~3–5 min for 25 subreddits
- **25 tracked subreddits** across 6 groups (Walmart core, Spark, Pharmacy, Sam's Club, International, Retail Competitors)
- **Incremental:** cursor-based ingestion — only new posts fetched per run

Request Sequence — End-to-End



Data Ingestion — Where the Plan Met Reality

How we replaced **PRAW** with the Arctic Shift API and turned a broken plan into a stable 60-minute ingestion loop

INITIALLY PLANNED

🐍 PRAW · Reddit Dev API

- Official Python client for Reddit
- OAuth-based, real-time listing & stream
- Canonical "correct" ingestion path

❌ **Blocker — could not secure working Reddit dev-API credentials in time (app-registration + auth-flow friction). Live streaming path parked.**

ACTUALLY SHIPPED

🧊 Arctic Shift API

- Public Reddit archive endpoint — no auth
- Windowed pulls across 25 Walmart-relevant subs
- Same fetcher interface — PRAW pluggable later

✅ **Mitigation — Arctic Shift keeps the pipeline unblocked. Zero downstream schema change; ingestion feeds the exact same `raw_posts` table.**

Side-by-Side

Dimension	🐍 PRAW (planned)	🧊 Arctic Shift (shipped)
Auth	Reddit app + OAuth token	None — public HTTP
Access model	Live listings / stream	Windowed archive queries
Historical reach	~1 000 posts / listing cap	Deep — years of history
Setup effort here	Blocked at approval	Working same day
Cost	Free · quota-bound	Free · community-hosted

🎯 Outcome for the pipeline

- **No slippage** — Stage 1 (Ingest) hits 25 subs · hourly in production.
- **Reversible choice** — `src/ingestion/` abstracts source; PRAW can slot in when credentials clear.
- **Bonus win** — Arctic's historical depth let us backfill the benchmark set (200 real posts) that ModernBERT trained on.

Next in series → Step 2: Preprocess · Step 3: Trust · Step 4: Analyze · Step 5: Aggregate · Step 6: Alerts

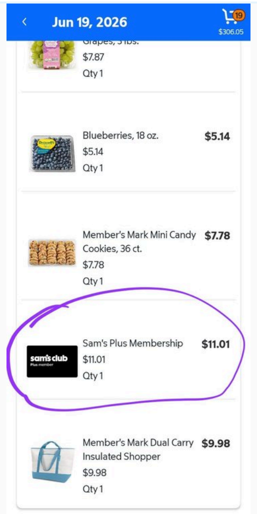
Vision Pipeline — Why Gemma 3 4B

3.9% of Reddit posts are image-only (empty body) — the complaint lives **inside the image**. Text-only pipeline drops these → we need a vision model to eliminate hallucination in multimodal retail analysis.

REAL REDDIT POST · R/SAMSCLUB id: 1uacdy7

Title: "Accidental Membership Upgrade"

↗ reddit.com/r/samsclub/comments/1uacdy7



Body: *(empty)* — the complaint is a purple circle drawn on the cart

❌ **TEXT-ONLY PIPELINE SEES**

Just the 3-word title. No item, no charge, no context. Sentiment ≈ neutral · Aspect = `unknown`. Silently dropped. **Useless.**

✅ **VISION (GEMMA 3 4B) EXTRACTS**

- **Sam's Club app cart**, Jun 19 2026 · total **$306.05**
- Circled · **Sam's Plus Membership $11.01** (purple ring = complaint anchor)
- Other items ignored: Blueberries $5.14, Mini Candy Cookies $7.78, Insulated Shopper $9.98
- Aspect = `fees / accidental charges` · Sentiment = **negative** · Conf **0.88**

🎯 **Signal recovered** — a UX complaint about an easy-to-tap upgrade toggle now feeds the *fees* aspect KPI and triggers alerts if the pattern repeats.

🧠 **VISION MODEL CHOSEN**

Gemma 3 4B

3.3 GB · Ollama · Google

DOCVQA · SIZE · RUNTIME

gemma3:4b	**83**	3.3 GB	✅ Ollama
LLaVA-1.6 8B	75	5.5 GB	larger, slower
PaliGemma 2 3B	81	~6 GB	no Ollama
LLaVA-1.5 7B	28	4.7 GB	3× worse OCR

WHY IT WINS

- Best DocVQA under 4 GB — reads receipts, app screens
- Google-maintained → Walmart policy compliant
- Reuses local Ollama (`localhost:11434`)
- 4–6 s warm latency per image

Vision Pipeline · Deep-Dive — 75% Failure → Root Cause → Multi-Pass Fix

First-cut Gemma 3 4B hallucinated on 1 in 2 images → root-caused → 5 papers reviewed (all China-blocked) → borrowed the *ideas* → 4-pass pipeline with a text-only final step.

⚠️ 1 · FIRST-CUT TEST · N = 8

baseline single-pass

75% overall fail (6/8) **50%** hallucination (4/8) **37.5%** critical (fake receipts/prices) **25%** correct (memes only)

Hallucinations seen: "receipt $39.99 w/ handwritten *Damaged Box*" (product page, no price, no handwriting) · "Order status PENDING" (word absent) · "12-pack Zero Sugar Dr Pepper" (single 42.3 fl oz regular bottle) → **fabricated data corrupted dashboards.**

🔍 2 · ROOT CAUSE (5)

- **768 px fixed resize** → small text unreadable
- **4 B params** → can't reason button + error + context
- **No dynamic resolution** → one fixed scale for whole image
- **Hallucinates under uncertainty** → invents rather than admits
- **No context awareness** → confuses screenshots vs physical displays

📚 3 · RESEARCH 2023–25

5 / 5 blocked

Walmart vendor policy

❌ **UReader** (Tencent) — shape-adaptive cropping ❌ **TextMonkey** (USTC) — shifted window attention ❌ **DocOwl 1.5** (Alibaba) — structure-aware parsing ❌ **InternVL2** (Shanghai AI Lab) — tile-based ❌ **Qwen2.5-VL** (Alibaba) — dynamic res + multimodal RoPE

Consensus: tile-based attention + dynamic/native resolution is THE fix. Fixed resize is the failure mode — *not* the model architecture.

🧭 4 · STRATEGY — TAKE THE IDEAS, NOT THE MODELS (SAME GEMMA3:4B, SMARTER CALLING)

Paper technique	We implement as
Tile-based processing (InternVL2)	Split image into 2–4 crops ourselves, send each to gemma3:4b
Structure-first parsing (DocOwl 1.5)	Pass 1 asks "what type of image?" before reading text
Focused text extraction (UReader)	"Read ALL text verbatim" prompt per tile
Resolution preservation (Qwen2.5-VL)	Tiling → 2–4× effective resolution without resizing

⚙️ 5 · MULTI-PASS PIPELINE (THE FIX)

PASS 1 · STRUCTURE
← DocOwl 1.5
"What TYPE?" → screenshot / photo / receipt / app / meme. Text-heavy → continue.

PASS 2 · TILE
← InternVL2
Split full image into 2–4 crops. Each crop = 2–4× effective resolution.

PASS 3 · EXTRACT
← UReader
Per tile: "Read ALL text verbatim" → quoted text, prices, errors, buttons.

PASS 4 · MERGE (no image)
text-only LLM call
Combine text observations into 2–4 sentences. Model **never sees image → cannot lie**.

Anti-hallucination insight: removing the image from the final generation step makes it *physically impossible* for the model to invent visual details — it can only work with text actually extracted in Pass 3.

Vision Pipeline · Results — Hallucination Eliminated

Initial 8-image validation

Metric	Before (Single-Pass)	After (Multi-Pass)	Change
Hallucination rate	50% (4/8)	**0% (0/8)**	**↓ 100%**
Overall failure rate	75% (6/8)	25% (2/8)	↓ 67%
Correct text extraction	25% (2/8)	75% (6/8)	**3× better**
Fabricated claims	8 total	**0**	Eliminated
Avg latency / image	~5 s	~15 s	3× (acceptable)

Scaled validation — 25 images

Verdict	Count	Percentage
✅ PASS (correct, no hallucination)	**22 / 25**	**88%**
⚠️ PARTIAL (correct but sparse)	3 / 25	12%
❌ FAIL (missed critical info)	**0 / 25**	**0%**

- Single-pass hallucinated on **44%** (11/25) of images at scale
- Multi-pass hallucinated on **0%** (0/25)
- **80%** of Walmart Reddit complaint images are screenshots/app screens — exactly the category where single-pass fails

Trust Score — Formula · 3 Components · Credibility Signals

How we validate post credibility. Each component ∈ **[0, 1]**, final score is clamped. Low-trust posts are **flagged for review, never dropped** (Req R5).

MASTER FORMULA

$$\text{trust\_score} = 0.4 \times \text{metadata} + 0.3 \times \text{dedup} + 0.3 \times \text{llm\_credibility}$$

1 · METADATA · W = 0.4

$\text{meta} = w_{base} + w_{age} \cdot age + w_{karma} \cdot karma + w_{len} \cdot len + w_{eng} \cdot eng$

Signal	Formula	w
Base floor	const	0.15
Account age	`min(age_d/365, 1)`	0.20
Karma	`min(karma/5000, 1)`	0.20
Length	`min((title+body)/200, 1)`	0.30
Engagement	`min(max(score,0)/20, 1)`	0.15

2 · DEDUP / ORIGINALITY · W = 0.3

- MD5 of normalized text (`lower + collapse ws`) vs rolling window of recent posts
- Frequency score: **1st** = 1.0 · **2nd** = 0.5 · **3–5** = 0.2 · **>5** = 0.0 (bot/copypasta)
- **Roadmap:** MiniLM-L6-v2 embeddings + cosine (`> 0.92` = near-dup) — dep in `requirements.txt`

3 · LLM CREDIBILITY · W = 0.3

- Invoked only when `0.3 < meta < 0.8` (ambiguous zone) — cost control
- Rule-based heuristic (free) *or* **gpt-4o-mini via WMT LLM Gateway**
- Checks: promotional language · URL stuffing · ALL-CAPS · retail insider terms

CREDIBILITY SIGNALS — SCORE IMPACT

❌ **Negative**

Promotional language (≥ 2 phrases)	**−0.25**
URL stuffing (≥ 3 links, short text)	**−0.20**
Karma/age mismatch (new acct, high karma)	**−0.20**
Excessive CAPS (> 40% letters)	**−0.15**
New account + promotional	**−0.20**

✅ **Positive**

Retail insider terms (≥ 2: OGP, ASM, CAP2, Spark...)	**+0.25**
Organic long-form (> 200 chars · no links · no promo)	**+0.10**
Single retail insider term	**+0.10**

ModernBERT — Why · 3-Stage Curriculum · Results

Thesis proven: long-context, domain-fine-tuned encoders beat short-context Twitter-trained baselines on Reddit-flavored retail complaints — **Macro F1 0.62 → 0.76 (+22%)**, long-post F1 **0.28 → 1.00**.

❌ ROBERTA LIMITS

- Only **512 tokens** → truncates long Reddit posts
- Trained on Twitter → different register from Reddit
- No domain knowledge of retail / Walmart terminology

✅ MODERNBERT WINS

- **8 192 tokens** (16× longer) — full posts, no truncation
- Modern arch: RoPE · GeGLU · alternating attention
- Fine-tunable with curriculum learning

🎓 3-STAGE CURRICULUM TRAINING

Stage	Dataset	Ep	Purpose
1 · Generic Sentiment	TweetEval-sentiment (45 K tweets)	2	Polarity grounding
2 · Reddit Register	GoEmotions-3class (54 K Reddit)	2	Reddit language
3 · Domain Special.	Walmart-200 (5-fold CV)	≤15*	Retail fine-tune

* patience = 3 · early stopping on `eval_macro_f1`

TRAINING CONFIG

- `max_length = 1024` (long-context lever)
- Effective batch **32** (BS 8 × grad-accum 4)
- Class weights `neg 0.52 · neu 1.03 · pos 8.33`
- Minority oversample to ~100/class/fold
- Apple M-series · MPS backend

🏆 FINAL RESULTS — 5-FOLD OOF CV

Metric	RoBERTa	ModernBERT v2	Δ
Macro F1 (overall)	0.6272	**0.7642**	**+0.137 (+22%)**
F1 negative	0.7967	0.8779	+0.081
F1 neutral	0.6087	0.7480	+0.139
F1 positive	0.4762	0.6667	+0.190
Long-post F1 (≥ 512 tok)	0.2778	**1.0000**	**+0.722** ✨
Short-post F1 (n = 193)	0.6360	0.7619	+0.126
Latency (ms/post, MPS)	6.5	11.9	+5.4

Long-context hypothesis proven: on posts ≥ 512 tokens, ModernBERT scores a perfect **1.00** vs RoBERTa's **0.28**.

ModernBERT · Training Evidence — Artifacts on Disk

> Proof that ModernBERT was actually fine-tuned in-house (not a downloaded checkpoint)

Training pipeline

Stage	Script / File	Output
Data collection	`scripts/fetch_real_benchmark.py`	`data/benchmark_real_200.jsonl` — 200 real Reddit posts, body 300–3604 chars
Human labeling	`scripts/label_benchmark.py` (interactive)	`human_sentiment` field per row (neg=127 · neu=65 · pos=8)
Curriculum trainer	`scripts/train_modernbert_sentiment.py`	3-stage checkpoints under `models/modernbert_walmart/`
Honest evaluation	`scripts/eval_sentiment_models.py`	`models/modernbert_walmart/eval_results.json`
Thesis chapter	`docs/MODEL_COMPARISON.md`	~290-line write-up of methodology + results

Checkpoints produced (on disk today)

```
models/modernbert_walmart/
├── stage1_tweeteval/      # after Stage 1  — TweetEval, macro F1 = 0.7267
├── stage2_goemotions/     # after Stage 2  — GoEmotions, macro F1 = 0.7028
├── stage3_walmart/        # 5-fold CV artefacts (cv_results.json, per-fold checkpoints)
├── final/                 # production checkpoint (max_length=1024, trained on all 200)
├── final_max512/          # v1 ablation (max_length=512) — kept for comparison
└── eval_results.json      # aggregated CV metrics + per-length-bucket F1
```

Reproduction command (offline)

```
HF_HUB_OFFLINE=1 TRANSFORMERS_OFFLINE=1 HF_DATASETS_OFFLINE=1 \
  /opt/miniconda3/bin/python scripts/train_modernbert_sentiment.py \
  --stages 1,2,3 --folds 5 --max-length 1024 --batch-size 8
```

Why not just use RoBERTa? — Decision matrix

Criterion	RoBERTa (`cardiffnlp/twitter-roberta-base`)	ModernBERT (`answerdotai/ModernBERT-base`)	Winner
Context length	512 tokens	**8192 tokens** (16×)	ModernBERT
Training corpus	Twitter (short, casual)	Web + code + long docs	ModernBERT
Long-post F1 (≥512 tok)	0.2778	**1.0000**	**ModernBERT**
Overall Macro F1	0.6272	**0.7642**	ModernBERT
Latency (MPS)	**6.5 ms**	11.9 ms	RoBERTa
Off-the-shelf quality	Strong baseline	Weak without fine-tuning	RoBERTa
Fine-tuning cost	Same	Same	Tie

Decision: ModernBERT wins on the metrics that matter (accuracy + long-context). We accept +5 ms latency for +0.137 macro F1. RoBERTa is retained as fallback in `config/models.yaml` if the local checkpoint is missing.

Trust · Confidence Score — Model Certainty

Confidence = softmax probability of the predicted class.

How it's calculated

1. ModernBERT outputs **logits** for `[negative, neutral, positive]`
2. **Softmax** converts to probabilities: P(neg), P(neu), P(pos)
3. `confidence = max(P(neg), P(neu), P(pos))`

Thresholds used in the system

Threshold	Value	Source
Analysis confidence	≥ 0.7	`config/models.yaml`
Notification P1	trust ≥ 0.70 **AND** confidence ≥ 0.80	dispatcher.py
Notification P2	trust ≥ 0.50 **AND** confidence ≥ 0.60	dispatcher.py

Combined Priority Formula

```
P1 = (trust_score ≥ 0.70) ∧ (confidence ≥ 0.80)   → immediate action
P2 = (trust_score ≥ 0.50) ∧ (confidence ≥ 0.60)   → review-worthy
< P2 → no notification triggered (still visible in dashboard)
```

> Dashboard displays **both** trust and confidence per post for analyst transparency.

Notification System — Architecture · Config Page · API

Group-based routing for P1 / P2 priority posts — email + Slack, per-team ownership

Priority classification & routing

- **P1** — `trust ≥ 0.70` AND `confidence ≥ 0.80` → immediate action
- **P2** — `trust ≥ 0.50` AND `confidence ≥ 0.60` → review-worthy
- Routing model: **per-subreddit-set** — different teams own different subreddits
- Each group has its own email DL, Slack channel, and priority filter (P1, P2, or both)

```
Pipeline → classify_priority(trust, conf)
  ├── Not P1/P2 → skip
  └── P1 / P2 → find matching notification groups
        ├── group matches subreddit + priority filter?
        │     ├── email DL configured → send email
        │     └── Slack channel configured → send Slack
        └── log to notification_log (audit trail)
```

Config UI at `/notifications` — admin workflow

- Create groups (name, subreddits, email DL, Slack)
- Quick-add subreddits by category (Walmart core, Spark, Pharmacy, International, Sam's, Competitors)
- Priority filter: P1, P2, or both per group
- Enable / disable via toggle
- **Test (dry-run)** — simulate notification without sending
- View delivery log — full audit trail of sent notifications
- Delete groups

API endpoints (8 total)

Method	Path	Purpose	Method	Path	Purpose
GET	`/api/notifications/config`	Overall config + groups	POST	`/api/notifications/groups`	Create new group
GET	`/api/notifications/groups`	List all groups	PUT	`/api/notifications/groups/{id}`	Update group
DELETE	`/api/notifications/groups/{id}`	Delete group	POST	`/api/notifications/test/{id}`	Dry-run test
GET	`/api/notifications/log`	Delivery audit	GET	`/api/notifications/subreddits`	Available subreddit list

Post Mid-Semester Work — Review · Explorer · Lifecycle · Insights · Results

All 5 areas built after mid-sem: human-in-the-loop workflows → competitor insights → consolidated metrics + tech stack

1 · Review & Validate

Analysts correct AI labels + draft replies.

- Queue sorted by priority (P1 first)
- Correct sentiment + aspects
- Click "Generate Drafts" → **dual draft**
  - **A** Smart Composer — deterministic phrase pool + keyword extraction
  - **B** Reply LLM — Mistral 7B (Ollama, primary) · FLAN-T5-base (HuggingFace fallback)
- Analyst picks → edit → post to Reddit
- **Learning loop**: replies → `feedback` table → few-shot examples for future drafts (tone matching improves without retraining)

2 · Post Explorer

Browse all analyzed posts with rich filtering.

- **Filters**: sentiment · confidence slider · trust slider · subreddit (25) · aspect (8-label) · date (today/week/month/custom) · full-text
- **Aspects**: pricing · product quality · customer service · store experience · online/app · delivery/pickup · returns · app_website
- **Card**: title + excerpt · sentiment badge · confidence % · trust indicator · aspect tags · subreddit · time
- **Actions**: Review · Add to Lifecycle · View Details · Reddit link

3 · Post Lifecycle (Kanban)

4-state board with 2-step resolve modal.

```
TRIAGED → ACKNOWLEDGED → IN PROGRESS → RESOLVED
new P1/P2   assigned      reply drafted  posted
```

- **Step 1**: save action note + LLM-drafted reply → "Save & open Reddit" (copies + opens thread) OR "Resolve (no reply)"
- **Step 2**: paste reply on Reddit → return → "Mark Resolved"
- All transitions timestamp-logged → SLA tracking

4 · Insights & Competitor

AI-generated strategic intelligence.

- **Issue Rankings** — volume × severity × recency, per aspect, trend arrows (up/down)
- **Competitor Pulse** — Walmart vs **Costco / Target / Amazon**, cross-mentioned posts, subreddit breakdown
- **LLM Summaries** — weekly themes, suggested action items, emerging-topic detection (new clusters in ≥5 posts / 2 hrs)
- **Aspect Drilldown** — 8-label taxonomy from `config/models.yaml` ; per-aspect sentiment trend + volume + word cloud + representative posts

5 · Results — Key achievements

Area	Achievement
ModernBERT	0.6272 → **0.7642** F1 (+22%)
Long-post	0.28 → **1.00** F1 (+722%)
Vision hallucination	50% → **0%**
Vision extraction	25% → **75%**
Pipeline	25 subs · hourly
Dashboard	7 pages · WS alerts
Notifications	Group-based P1/P2
Lifecycle	Full Kanban + 2-step

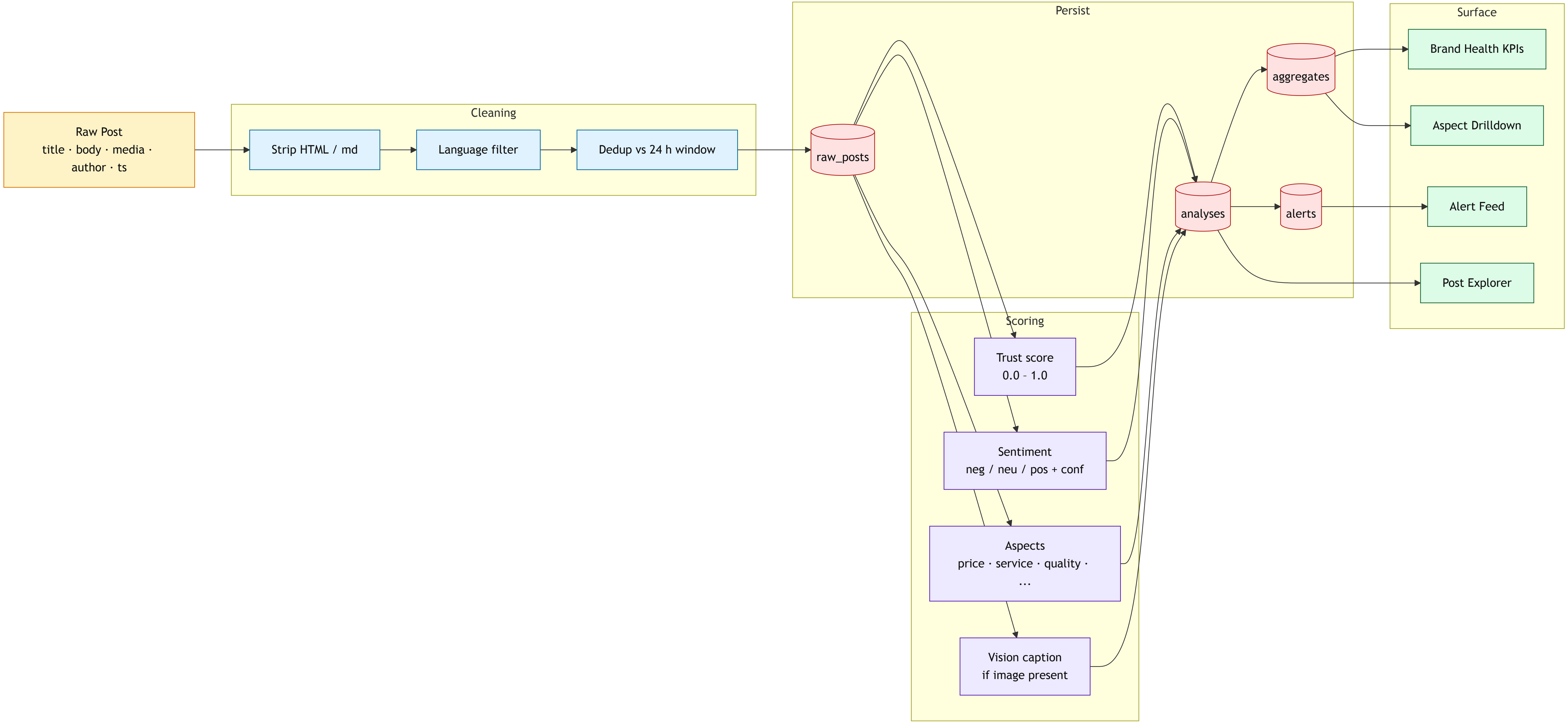
Evidence: 5-fold OOF CV · 7 posts ≥512 tok · 8+25 img validation

Technical stack summary

Layer	Technology	Key decision
Backend	Python 3.13 · FastAPI · SQLite	Free · local-first · modular
Frontend	React 18 · TS · Vite · Tailwind	Modern SPA · responsive
Sentiment	ModernBERT (fine-tuned · 1024-tok)	Domain-specialized · offline
Aspects	DeBERTa-v3 zero-shot-v2 (BART-MNLI fallback)	No training needed
Vision	Gemma 3 4B via Ollama (multi-pass)	Policy compliant · no hallucination
Reply Gen	Mistral 7B (Ollama) + FLAN-T5-base (HF fallback) + Smart Composer	Dual draft · local-first · learning loop
Trust LLM	gpt-4o-mini via WMT LLM Gateway (ambiguous-zone only)	Cost-controlled cloud call · rule-based fallback
Trust · Scheduling · Observability	Metadata+Dedup+LLM · APScheduler 60 min + manual · structlog + JSONL cost ledger	Flag-don't-drop · cursor-based · per-call cost

Data Flow — From Raw Post to Dashboard KPI

How a single Reddit post transforms into a scored, stored, and surfaced artefact



FINAL TAKEAWAY

From Reddit Noise to Decision-Ready Retail Intelligence

RSI turns raw community chatter into trusted signals for Ops, PR, and Product through a local-first AI pipeline, credibility gating, and human-in-the-loop review.

WHY THIS SYSTEM IS DIFFERENT

TRUST FIRST

Metadata + originality + credibility scoring reduce noisy or manipulative posts before they affect business decisions.

RETAIL AWARE

ModernBERT, DeBERTa-v3, and Gemma 3 capture long complaints, aspect-level issues, and receipt or screenshot evidence.

HUMAN CONTROLLED

Analysts validate labels, choose replies, and close the loop so the system learns without unsafe auto-posting.

OPERATIONALLY USEFUL

P1/P2 routing, dashboards, lifecycle tracking, and group-based alerts convert model output into team action.

SENTIMENT UPLIFT

+22%

Macro F1 improved from 0.6272 to 0.7642

VISION RELIABILITY

0%

Hallucination after switching to Gemma 3 multi-pass analysis

COVERAGE

25

Retail communities monitored hourly with live aggregation

ACTION LOOP

P1 / P2

Prioritized alerts, analyst review, and lifecycle closure

CLOSING MESSAGE

PIPELINE ARC